

A Base-Space Obstruction for Continuous $C_0(X)$ -Algebras with BCP

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June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let X be a locally compact Hausdorff space. Recall that a $C_0(X)$ -algebra is a C^* -algebra A equipped with a nondegenerate homomorphism $C_0(X) \rightarrow ZM(A)$. For $x \in X$, write A_x for the fiber and $a(x)$ for the image of $a \in A$ in A_x . We call A continuous when $x \mapsto \|a(x)\|$ is continuous for every $a \in A$. We use the standard fiber norm formula

$$\|a\| = \sup_{x \in X} \|a(x)\|$$

for $C_0(X)$ -algebras.

Theorem 1. *Let A be a continuous $C_0(X)$ -algebra and assume that $A_x \neq 0$ for every $x \in X$. If A has the ball-covering property, then X has a countable π -basis.*

Corollary 1. *If X has no countable π -basis and A is a continuous $C_0(X)$ -algebra with nonzero fibers over every point of X , then A fails BCP.*

Idea of the Proof

The obstruction from $C_0(K)$ and $C_0(K, E)$ is not tied to trivial bundles. Given countably many candidate centers a_n , continuity of the fiber-norm functions $x \mapsto \|a_n(x)\|$ gives countably many open near-norm sets. If the base has no countable π -basis, some nonempty open set W hides from all these near-norm sets. A norm-one element supported inside W vanishes at points outside W where each center a_n almost attains its norm. Therefore it escapes every ball $B(a_n, \|a_n\|)$, so no countable BCP cover can exist.

Proof

We use the usual equivalent form of BCP: a normed space Y has BCP iff there are countably many nonzero $y_n \in Y$ such that

$$S_Y \subseteq \bigcup_{n=1}^{\infty} B(y_n, \|y_n\|).$$

Indeed, every open ball off the origin has radius at most the norm of its center, and enlarging to radius $\|y_n\|$ still leaves the origin outside the open ball.

Lemma 1. *If A is a $C_0(X)$ -algebra with $A_x \neq 0$ for all $x \in X$, then for every nonempty open set $W \subset X$ there is $b \in A$ with $\|b\| = 1$ and $b(x) = 0$ for all $x \notin W$.*

Proof. Choose $x_0 \in W$. Since $A_{x_0} \neq 0$, there is $a \in A$ with $a(x_0) \neq 0$. By local compactness and regularity, choose $h \in C_c(X)$ with $0 \leq h \leq 1$, $h(x_0) = 1$, and $\text{supp } h \subset W$. Then $ha \neq 0$, and $(ha)(x) = 0$ for $x \notin W$. Normalize $b = ha/\|ha\|$. $\square$

Proof of Theorem 1. We prove the contrapositive. Assume that X has no countable π -basis. Let (a_n) be any sequence in A . We shall find $b \in S_A$ such that

$$\|b - a_n\| \geq \|a_n\|, \quad n \geq 1.$$

Thus the balls $B(a_n, \|a_n\|)$ do not cover the unit sphere.

For $n, k \geq 1$, define

$$U_{n,k} = \{x \in X : \|a_n(x)\| > \|a_n\| - 1/k\}.$$

Whenever $a_n \neq 0$, these sets are nonempty and open: nonemptiness follows from the fiber norm formula, and openness follows from continuity of $x \mapsto \|a_n(x)\|$. Since X has no countable π -basis, there is a nonempty open set $W \subset X$ such that no nonempty $U_{n,k}$ is contained in W .

By Lemma 1, choose $b \in S_A$ with $b(x) = 0$ for all $x \notin W$. Fix n . If $a_n = 0$, then $\|b - a_n\| = 1 \geq \|a_n\|$. Otherwise, for each k choose

$$x_{n,k} \in U_{n,k} \setminus W.$$

At this point $b(x_{n,k}) = 0$, hence

$$\|b - a_n\| \geq \|(b - a_n)(x_{n,k})\| = \|a_n(x_{n,k})\| > \|a_n\| - 1/k.$$

Letting $k \rightarrow \infty$ gives $\|b - a_n\| \geq \|a_n\|$. This holds for every n , so the sequence (a_n) cannot be a BCP family. As (a_n) was arbitrary, A fails BCP. $\square$

Corollary 1 is the contrapositive form just proved.

Relation to the Full C^* -Algebra Question

The theorem is a genuine continuous-field extension of the negative half of the $C_0(K, E)$ argument. It shows that every visible continuous central base of a C^* -algebra with BCP must have countable π -basis.

It is not a full characterization of arbitrary C^* -algebras. The positive direction for nontrivial fields requires coherent local directions across fibers. A countable π -basis of the base only localizes where a section is large; it does not give a countable set of fiber directions that can be used to build centers. This is precisely where the trivial-bundle proof uses constant fiber vectors.

The continuity hypothesis is also real. For a merely upper semicontinuous $C_0(X)$ -algebra, the superlevel sets $\{x : \|a(x)\| > \|a\| - \varepsilon\}$ need not be open, so the π -basis argument above does not apply without an additional lower-semicontinuity or openness assumption.

Literature and Search Bounds

The run indexes were searched for arXiv:2311.14257, C^* -algebras, BCP, UBCP, pure state space, π -basis, $C_0(X)$ -algebras, continuous fields, and upper semicontinuous fields. Web searches on June 17, 2026 found the source paper [1], the vector-valued $C_0(K, X)$ theorem of Liu–Liu–Lu–Zheng [2], the Calkin-algebra obstruction of Bao–Liu–Shen [3], and the later operator-space approximation criteria [4]. No direct characterization of arbitrary C^* -algebras with BCP/UBCP was found.

Human Review Focus

The proof is short. The main points to verify are:

- the standard continuous $C_0(X)$ -algebra fiber norm facts used at the start;
- Lemma 1, especially that nonzero fibers give nonzero local elements in every nonempty open set;
- the final escape estimate, where the local element vanishes outside W at points where each candidate center is almost norm-attaining.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Qiyao Bao, Rui Liu, and Jie Shen, *The ball-covering property of non-commutative spaces of operators on Banach spaces*, arXiv:2412.07137, 2024.
- [4] Qiyao Bao, Rui Liu, and Jie Shen, *Approximation properties and quantitative estimation for uniform ball-covering property of operator spaces*, arXiv:2507.02261, 2025.